A PROJECT MOON PI PROJECT · USER GUIDE

SkyScope

A live aircraft radar styled after an air-traffic-control display, for a Raspberry Pi on a TV or monitor, or any web browser.



SkyScope centred on Phoenix Sky Harbor at 25 miles: Class B airspace, runways with extended centrelines, arrival routes (amber), aircraft coloured by kind, and the information panel.

Project: github.com/ricsne2010/skyscope · Licence: MIT · Not for navigation.

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1. What SkyScope does

SkyScope shows the aircraft around a location you choose, live, on a radar scope. Each aircraft is drawn at its position with a symbol shaped like the aircraft (airliner, business jet, light aircraft, helicopter…), a line showing where it will be in a minute, and a label with its callsign, type and altitude. A side panel lists every aircraft in range with its altitude, distance, bearing and speed, and a card shows details and a photo of the nearest or selected aircraft.

Underneath the traffic you can show an air-traffic-control style map: airports and runways with extended centrelines, US Class B, C and D airspace, and the published departure (SID) and arrival (STAR) routes for airports you pick.

Two ways to display it

Display	Best for	How
Pi display	A Raspberry Pi connected to a TV or monitor, running full screen by itself.	`python3 skyscope.py`
Web radar	Any computer, tablet or phone on your network, or a laptop on its own. Sharp on high-resolution screens; extra features such as clicking aircraft and the filter panel.	Open `http://<address>:7700/radar`, or run `python3 skyscope.py --web-only` on a laptop.

Both run at the same time: a Pi can drive the TV while anyone on the network opens the web radar. Everything is configured from a **settings web page** at `http://<address>:7700/radar_config`, so the Pi needs no keyboard once it's set up.

Where the aircraft data comes from

- **Internet (default):** the free adsb.fi service, which pools reports from thousands of volunteer receivers. Needs an internet connection.
- **Your own receiver (optional):** a USB radio receiver on the Pi picks up aircraft signals directly, with no internet needed. See section 11.

> **Not for navigation.** SkyScope is for watching traffic. Positions can be delayed or missing, and map layers are approximations.

2. Requirements

Raspberry Pi setup

Item	Needed	Notes
Raspberry Pi	Pi 5 (recommended) or Pi 4 with 2 GB+ memory	Older models may work but will be slow at long ranges.
Power supply	Official supply for your model	Pi 5: 27 W USB-C. Pi 4: 15 W USB-C.
microSD card	16 GB or larger	Class A1 or better.
Display	Any HDMI TV or monitor	Widescreen (16:9) is ideal; other shapes work too. A micro-HDMI cable for Pi 4/5.
Network	Wi-Fi or Ethernet	Required for internet data, airline logos and photos.
Keyboard and mouse	For installation only	Afterwards everything is done from the settings page.
Software	Raspberry Pi OS with desktop (64-bit)	Includes Python 3. SkyScope also needs `pygame` and `git` (installed below).

Optional extras

Item	What it adds
USB SDR receiver (RTL-SDR) and a 1090 MHz antenna	Receive aircraft yourself, with no internet. A dongle with a built-in 1090 MHz filter and amplifier (sold for ADS-B) works best. Uses the free `readsb` software.
USB GPS receiver	Centres the radar on the Pi's own position, once or continuously (for a mobile setup). Uses the standard `gpsd` service.

Laptop or desktop instead of a Pi

- macOS, Windows or Linux with **Python 3.10 or newer**.
- The `pygame` package (`pip3 install pygame`).

- Any current web browser (Chrome, Safari, Firefox, Edge).

Network access used

SkyScope serves its pages on **TCP port 7700** on your local network. It connects out (HTTPS) to `opendata.adsb.fi` for aircraft, `api.planespotters.net` for photos, and `raw.githubusercontent.com` for airline logos.

3. Installing on a Raspberry Pi

1. **Prepare the SD card.** On another computer, install Raspberry Pi Imager. Choose your Pi model, **Raspberry Pi OS (64-bit)** (the version *with* desktop), and your SD card. When asked to customise settings, set:
   - **Hostname:** `skyscope` (so the settings page is at `http://skyscope.local:7700`)
   - **Username and password** of your choice
   - **Wi-Fi** network and password (skip if using Ethernet), and your time zone

   Write the card, put it in the Pi, connect the display, keyboard and mouse, and power on.

2. **Open a terminal** on the Pi (the black icon in the top bar) and bring the system up to date:

```
sudo apt update
sudo apt full-upgrade -y
```

3. **Install what SkyScope needs:**

```
sudo apt install -y python3-pygame git
```

4. **Download SkyScope** into your home folder:

```
git clone https://github.com/ricsne2010/skyscope.git ~/skyscope
```

5. **Start it:**

```
python3 ~/skyscope/skyscope.py
```

   The radar fills the screen. Press `Q` to quit. The first start creates `~/skyscope/settings.json`, which holds your settings.

6. **Start it automatically when the Pi boots.** This copies the included launcher into your autostart folder, pointing it at your home folder:

```
mkdir -p ~/.config/autostart
sed "s#/home/pi#$HOME#" ~/skyscope/skyscope.desktop > ~/.config/autostart/skysc
```

Restart the Pi (`sudo reboot`); SkyScope should open full screen after the desktop appears.

> If it doesn't start on its own, add it to the desktop's own autostart list instead:
>
> ```
> mkdir -p ~/.config/labwc
> echo "python3 $HOME/skyscope/skyscope.py &" >> ~/.config/labwc/autostart
> ```

7. **Set it up from another device.** On a phone or computer on the same network, open `http://skyscope.local:7700/radar_config` (or `http://<pi-address>:7700/radar_config` ; the address is shown at the bottom of the Pi's screen) and continue with first-time setup.

Updating

```
cd ~/skyscope && git pull
```

Then restart SkyScope (or the Pi). Your `settings.json` is kept.

If the Pi struggles

Very long ranges (150 miles and beyond) near busy areas can mean 500+ aircraft. If the Pi display becomes jerky: use filters to show fewer aircraft, choose a shorter range, set the Pi's screen to 1280×720 (Screen Configuration in the Pi menu), or lower `fps` in `settings.json` . Another option is to show the web radar full screen in the Pi's browser instead of the Pi display:

```
python3 ~/skyscope/skyscope.py --web-only &
chromium --kiosk http://localhost:7700/radar
```

4. Running on a laptop or desktop

SkyScope runs on a Mac or a Windows 10/11 PC as well as a Pi. On a computer you use the web radar in your browser, and a desktop icon with a radar-screen picture starts it with one click.

Mac

1. Install Python 3.10 or newer from python.org (open the downloaded installer and follow it).
2. Download SkyScope: on its GitHub page choose **Code → Download ZIP**, and unzip it somewhere it can stay, such as your Documents folder. (Or `git clone https://github.com/ricsne2010/skyscope.git`.)
3. Open **Terminal**, type `cd ` (with a space), drag the SkyScope folder onto the Terminal window, and press Return. Then install pygame and make the desktop icon:

   ```
   python3 -m pip install pygame
   sh tools/make_mac_app.sh
   ```

4. Double-click the new **SkyScope** icon on your Desktop. The radar opens in your browser after a few seconds. If macOS asks whether Python may accept incoming network connections, choose **Allow**, so phones and Satellite displays can reach it.

Windows 10 or 11

1. Install Python 3.10 or newer from python.org. On the installer's first screen, tick **Add python.exe to PATH**, then choose **Install Now**.
2. Download SkyScope: on its GitHub page choose **Code → Download ZIP**. Right-click the ZIP, choose **Extract All**, and put the folder somewhere it can stay, such as Documents.
3. In the SkyScope folder, open **tools**, then **windows**, and double-click **create-shortcut.bat**. If Windows warns about a file from the internet, choose **More info → Run anyway**. The window installs pygame if needed and makes the shortcuts. Press a key to close it when it says *Done*.
4. Double-click the new **SkyScope** icon on your Desktop (it's also in the Start menu). The radar opens in your browser after a few seconds. If Windows Firewall asks about Python, allow it on **private networks**.

Using the icon

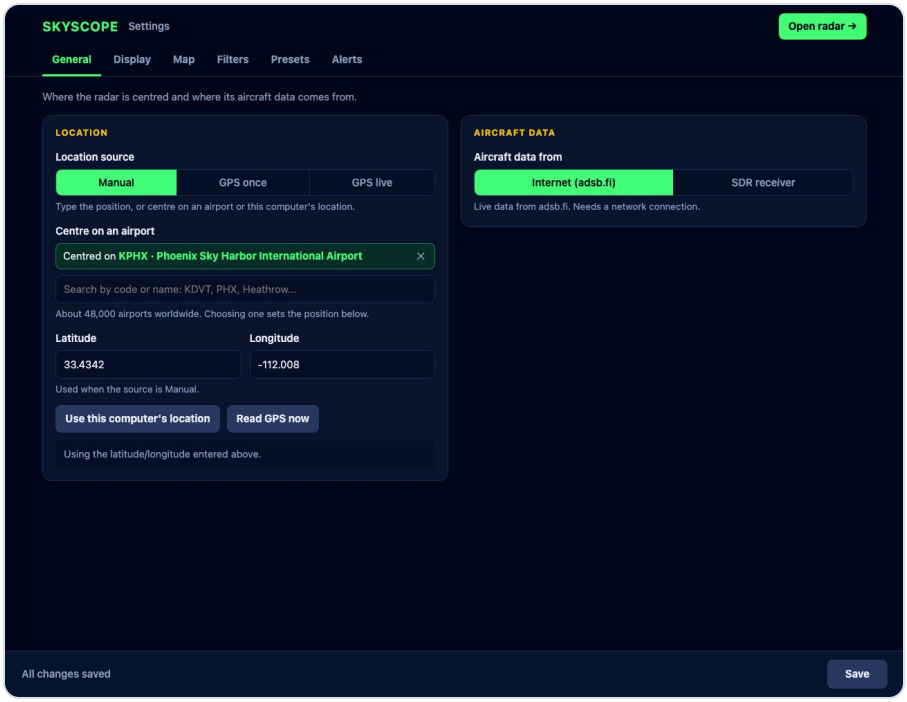
- Click it to start SkyScope. It runs in the background, with no window of its own, and opens the radar. Clicking it while SkyScope is running just opens the radar again.
- After restarting the computer, click it again. SkyScope keeps all your settings.
- To stop SkyScope, quit its **Python** process in Activity Monitor (Mac) or Task Manager (Windows).
- If it doesn't start, a message says so, and the details are in `skyscope.log` in the SkyScope folder. If you move the SkyScope folder or reinstall Python, make the icon again (step 3).

> **Without the icon:** in a terminal, in the SkyScope folder, run `python3 skyscope.py --web-only` (on Windows `py skyscope.py --web-only`), and stop it with `Ctrl`+`C`. To use the Pi-style display in a window instead: `python3 skyscope.py --window 1280x720`. The web radar is sharper on high-resolution (Retina or scaled) laptop screens.

> **Newest Python and pygame:** a brand-new Python version sometimes has no ready-made pygame for a while. If installing pygame fails, install the previous Python version (for example 3.13) and try again.

5. First-time setup

Open the settings page (`http://<address>:7700/radar_config`). The tabs across the top hold every option; nothing changes until you press **Save** in the bar at the bottom, which shows "Unsaved changes" whenever there's something to save.



The General tab: where the radar is centred, and where its aircraft data comes from.

1. **Choose the centre.** On the **General** tab, with **Manual** selected, either search for an airport (by code or name, such as `KDVT`, `PHX` or `Heathrow`) and pick it, or type a latitude and longitude. On a laptop, **Use this computer's location** fills them in for you.
2. **Pick your units** on the **Display** tab: distance (km, statute miles or nautical miles), speed and altitude.
3. **Pick a range** on the Display tab: the distance from the centre to the outer ring.
4. Press **Save**, then **Open radar** (top right) to see the result.

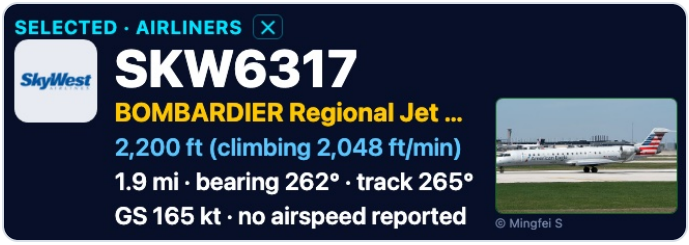
6. Using the radar

The scope

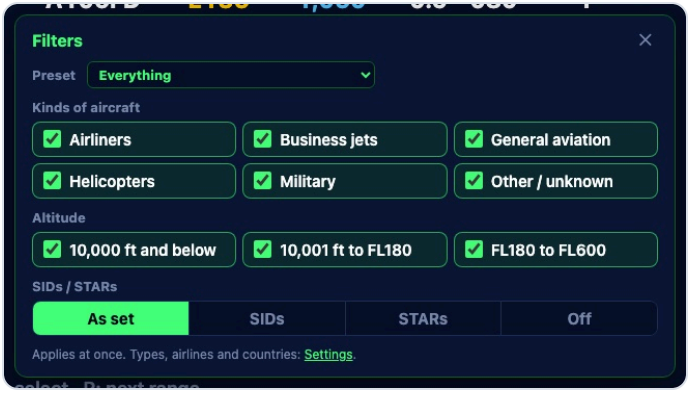
- **Range rings** are labelled with their distances along the east line. The range is the distance to the outer ring.
- **Aircraft** are drawn with a shape for their kind and a line showing where they'll be in about a minute. Labels give callsign, type and altitude; in busy areas some labels are left off to keep things readable (every aircraft is still in the list).
- The **yellow circle** marks the nearest aircraft. **Gold rings** mark watched aircraft and a **flashing red** symbol marks an emergency (see section 10).
- The green **sweep** is decorative; its speed and trail can be changed or switched off.
- Optional **dots on the rim** show aircraft just outside the range, in their direction.

The panel

The top shows the centre, the number of aircraft, the data source and how fresh the data is. The **card** below shows the nearest aircraft, or the one you've selected: callsign, aircraft type and registration, altitude and climb or descent, distance and bearing, and ground speed (plus airspeed when the aircraft reports it). When Planespotters.net has a photo of that exact aircraft, a thumbnail appears; click it to open the full photo. The **list** shows every aircraft in range, nearest first, with units under each column heading.



A selected aircraft, with its airline, type, registration and a photo of that exact aircraft.



The web radar's Filters panel: kinds of aircraft, altitude blocks, presets and SIDs/STARs.

On the web radar

- **Click an aircraft** (its symbol or label) or a **row in the list** to select it. Brackets mark it on the scope and the card shows its details. Click it again, or press `Esc`, to clear the selection. Hovering over a row highlights that aircraft on the scope.
- ⚙ **Filters** at the bottom of the panel opens a panel to choose kinds of aircraft, altitude blocks, a preset and the SIDs/STARs switch. Changes apply at once.
- `R` or `Space`: next range. `F`: full screen. Clicking empty space never changes the range.
- **Settings** and **Guide** links at the bottom open the settings page and this guide.

On the Pi display

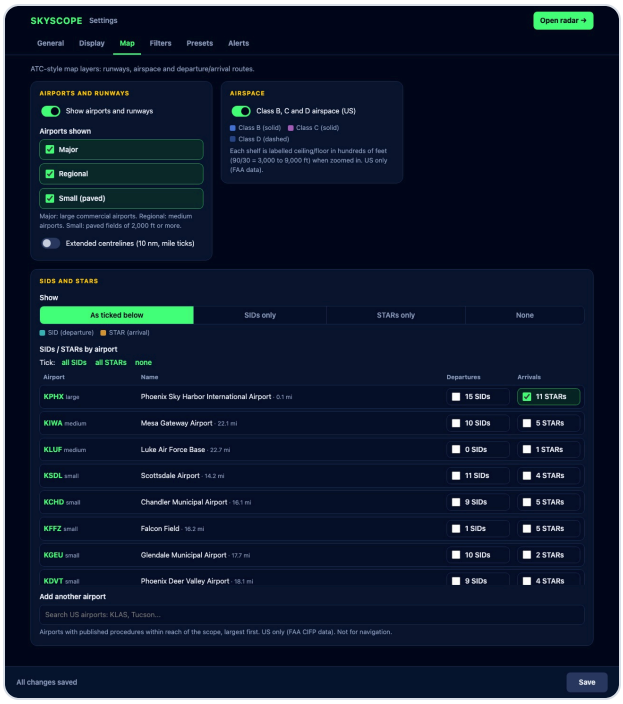
With a keyboard attached: `R` / `Space` / `→` next range, `←` previous range, `U` distance units, `W` runways, `S` sweep, `A` symbol style, `L` logos, `D` rim dots, `G` aircraft on the ground, `F` full screen, `Q` quit. Everything else is on the settings page.

7. The settings page

Tab	What's on it
General	Where the radar is centred (manual, an airport, or GPS) and where aircraft data comes from (internet or your own receiver).
Display	Range; aircraft symbols (silhouettes or simple triangles); colouring aircraft by kind; airline logos; the aircraft photo; rim dots; aircraft on the ground; units; the sweep.
Map	Airports and runways, extended centrelines, airspace, and SIDs/STARs per airport. See section 8.
Filters	Which aircraft appear. See section 9.
Presets	Named sets of filters: apply, save and delete.
Alerts	Emergency highlighting and the watchlist. See section 10.
About	SkyScope's version, the addresses of this radar (including the one to give a Satellite display), links to this guide and the project pages, data sources and credits.

The page works on phones as well as computers. Anyone on your network can open it; there is no password.

8. The ATC map



The Map tab: airport types, centrelines, airspace, and SIDs/STARs chosen per airport.

Airports and runways

Runways are drawn to scale. Runways of 5,000 ft or more get **extended centrelines** reaching 10 nautical miles out, with a tick every mile and a longer tick every five, as on an approach-control display. Runway numbers appear at the ends when zoomed in. Choose which airports appear: **Major**, **Regional** and **Small (paved)**, each on or off. Airport data covers the whole world.

Airspace (US)

Class B (solid blue), Class C (solid magenta) and Class D (dashed blue) outlines from the FAA. When zoomed in, each shelf is labelled with its ceiling over its floor in hundreds of feet: `90/30` means from 3,000 to 9,000 feet.

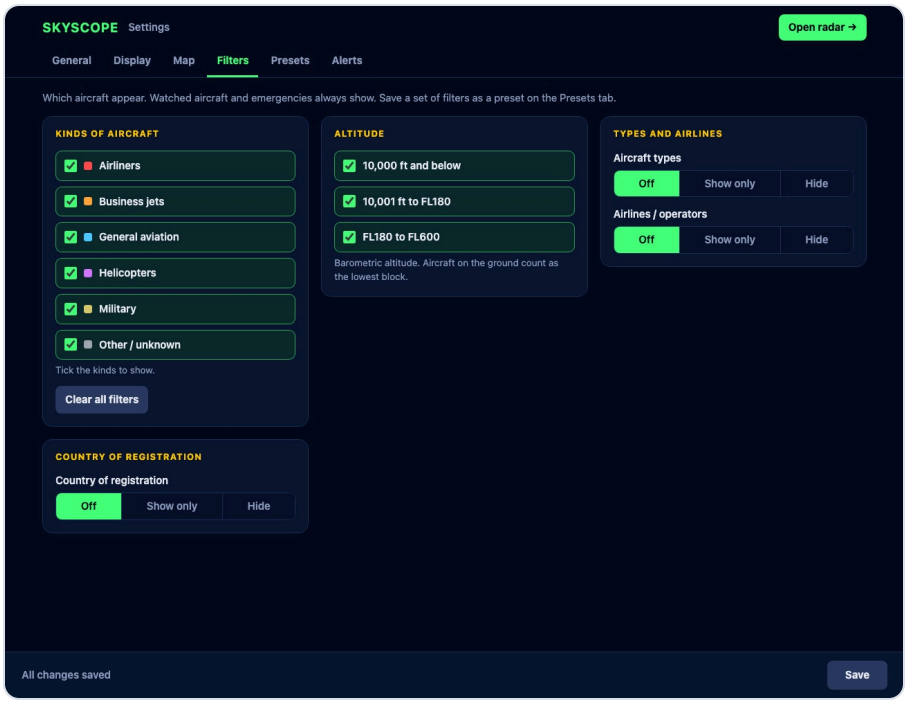
SIDs and STARs (US)

The Map tab lists every airport with published departure (SID) and arrival (STAR) routes within reach of the scope, larger airports first, with how many each has. Tick **SIDs**, **STARs** or both for each airport you want; use **Add another airport** for one further away. Departures are drawn in teal

and arrivals in amber, with the waypoints (△) named. The **Show** switch (also in the web radar's Filters panel) hides them all, or shows only SIDs or only STARs, without losing your choices.

> Parts of some routes are flown on a heading with no fixed track ("climb on heading 078, expect vectors"). Those parts can't be drawn, so a route may start or end away from the runway.

9. Filters and presets



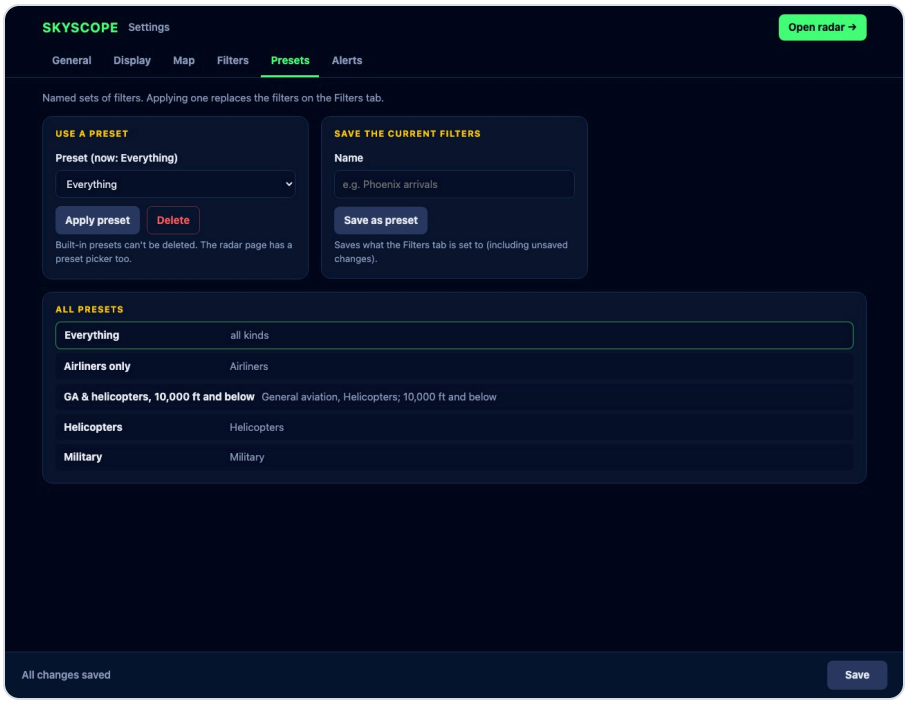
The Filters tab. Tick what you want to see.

- **Kinds of aircraft:** airliners, business jets, general aviation, helicopters, military, and other/unknown. A light aircraft flying a flight school's callsign counts as general aviation.
- **Altitude:** 10,000 ft and below; 10,001 ft to FL180; FL180 to FL600.
- **Aircraft types:** show only, or hide, a list of type codes such as `B738, A32*, C172` (`*` matches anything).
- **Airlines:** show only, or hide, by the three-letter code at the start of the callsign, such as `AAL, SWA, UAL`.
- **Country of registration:** worked out from each aircraft's transponder address, so it works even when the registration is unknown.

While any filter is on, the panel says, for example, "35 of 134 aircraft (filtered)". Emergencies and watched aircraft are always shown.

Presets

Built-in presets: *Everything*, *Airliners only*, *GA & helicopters, 10,000 ft and below*, *Helicopters* and *Military*. To make your own, set the filters you want, then on the **Presets** tab type a name and press **Save as preset**. Apply presets on the Presets tab or from the web radar's Filters panel.



The Presets tab lists every preset with a summary of what it shows.

10. Alerts and the watchlist

- **Emergencies:** aircraft squawking 7700 (emergency), 7600 (radio failure) or 7500 (hijack), or reporting an emergency status, flash red with a pulsing ring, are labelled first, move to the top of the list and get a red banner in the panel.
- **Watchlist** (Alerts tab): callsigns, registrations, transponder addresses or type codes, with `*` wildcards, for example `N123AB, ENY*, B52`. Matching aircraft get a gold ring and a ★ in the list, are labelled first, and optionally get a gold banner while in range.

11. Your own receiver and GPS

Receiving aircraft yourself

1. Plug an RTL-SDR dongle with a 1090 MHz antenna into the Pi. Place the antenna as high as possible with a clear view of the sky.

2. Install `readsb`, the decoding software, and tell it your location:

```
sudo bash -c "$(wget -O - https://github.com/wiedehopf/adsb-scripts/raw/master/
sudo readsb-set-location <your-latitude> <your-longitude>
```

3. On the settings page's **General** tab, set **Aircraft data from** to **SDR receiver** and save. With the location left at `auto`, SkyScope finds readsb (or dump1090/PiAware) by itself; the status line says which it's reading.

Range now depends on your antenna and its view of the sky. Aircraft types and names may be blank for some aircraft unless readsb's aircraft database is enabled.

Centring on a GPS position

1. Plug in a USB GPS receiver and install the GPS service:

```
sudo apt install -y gpsd
```

   USB receivers are usually detected automatically. For a GPS add-on board on the serial port, set `DEVICES="/dev/serial0"` in `/etc/default/gpsd` and run `sudo systemctl restart gpsd`.

2. On the **General** tab, choose **GPS once** (reads the position when SkyScope starts: for a fixed installation) or **GPS live** (keeps re-centring every few seconds as you move), and save. The status line shows the fix and number of satellites.

12. Troubleshooting

Problem	What to try
The panel says "No data" or "Waiting for data"	Check the Pi's internet connection. With an SDR receiver, read the status line on the General tab: it says whether readsb was found and whether its data is current.
"HTTP Error 429: Too Many Requests"	adsb.fi allows one request per second. Make sure only one copy of SkyScope runs per network address; it recovers by itself.
The settings page won't open from another device	Use the address shown at the bottom of the Pi's screen. Both devices must be on the same network. `.local` names can be slow on some devices; use the numeric address instead.
An aircraft is in the list but not on the scope	Aircraft are drawn once they're inside the outer ring; the list shows the same set. Turn on rim dots (Display tab) to see aircraft just beyond the range.
No photo for an aircraft	Photos only appear when Planespotters.net has one of that exact aircraft; many light aircraft and helicopters have none.
No SIDs or STARs appear	Tick SIDs or STARs for at least one airport on the Map tab, and check the Show switch isn't set to None. Only US airports have this data.
"Use this computer's location" is missing	Browsers only share location with the computer running SkyScope (`localhost`). On a Pi, type the position, pick an airport, or use GPS.
The Pi display is jerky	See If the Pi struggles in section 3.

13. Quick reference

Address	What it is
`http://<address>:7700/radar`	The web radar (also the bare `:7700` address)
`http://<address>:7700/radar_config`	The settings page
`http://<address>:7700/radar_config#about`	The settings page's About tab
`http://<address>:7700/guide`	This guide
`http://<address>:7700/api/satellite`	The feed SkyScope Satellite displays read

Command	What it does
`python3 skyscope.py`	Pi display, full screen
`python3 skyscope.py --window 1280x720`	Pi display in a window
`python3 skyscope.py --web-only`	Web radar only, and opens it in the browser

Ranges	Choices (distance to the outer ring)
Statute miles	2, 5, 10, 15, 25, 50, 75, 100, 150, 200, 250
Kilometres	5, 10, 15, 25, 50, 100, 150, 200, 300, 400
Nautical miles	2, 5, 10, 20, 40, 60, 80, 100, 150, 200

All settings are stored in `settings.json` in the SkyScope folder. The README lists every entry for anyone who prefers to edit it by hand.

14. SkyScope Satellite and Mini

Two smaller companions run on an inexpensive 4.3-inch touch display with a built-in ESP32-S3 microcontroller (the Sunton ESP32-8048S043, 800 × 480). Each has its own project page with installation instructions.

	SkyScope Satellite	SkyScope Mini
What it is	A second screen for this SkyScope: it shows the same radar, range, filters and map	A small radar of its own; no Raspberry Pi needed
Data	From this SkyScope over your network	From adsb.fi over Wi-Fi
Map	Everything this SkyScope shows: runways, airspace, SIDs and STARs	Runways at major airports
Set up from	This settings page (it follows SkyScope's settings)	Its own small web page
Project page	github.com/ricsne2010/skyscope-satellite	github.com/ricsne2010/skyscope-mini

Connecting a Satellite

A Satellite looks for `skyscope.local` on your network unless told otherwise. Either name the SkyScope computer `skyscope` (on a Raspberry Pi: `sudo raspi-config` → System Options → Hostname), or type the address shown under **Satellite address** on the **About** tab into the Satellite's setup page. SkyScope must be running with the settings page on (port 7700, the default). Several Satellites can follow one SkyScope.

15. Credits and data sources

SkyScope grew out of ESP32-Plane-Radar by MatixYo (MIT licence), a radar for a small round display on an ESP32 microcontroller.

Data	Source	Terms
Live aircraft	adsb.fi open data, or your own receiver	adsb.fi's terms (one request per second)
Airports and runways	OurAirports	Public domain
US airspace	FAA Class Airspace data	Public domain
US SIDs and STARs	FAA CIFP (updated every 28 days)	Public domain
Country from transponder address	ICAO address allocations	Facts
Airline logos	Jxck-S/airline-logos, downloaded when first needed	Airlines' trademarks
Aircraft photos	Planespotters.net, shown with the photographer's credit	Photographers' copyright

> **Not for navigation.** Aircraft positions may be delayed, incomplete or wrong, and map layers are simplified. Never use SkyScope for flight planning or navigation.



SkyScope is a Project Moon Pi project and free software under the MIT licence. Source: github.com/ricsne2010/skyscope.